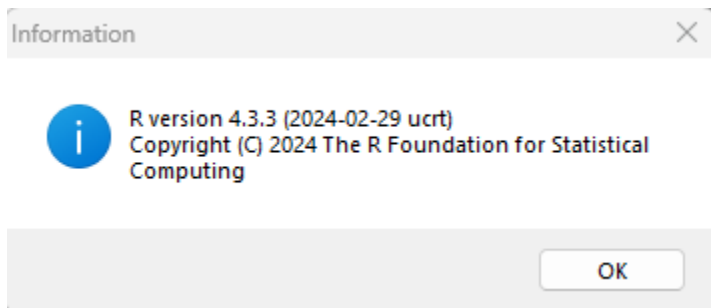


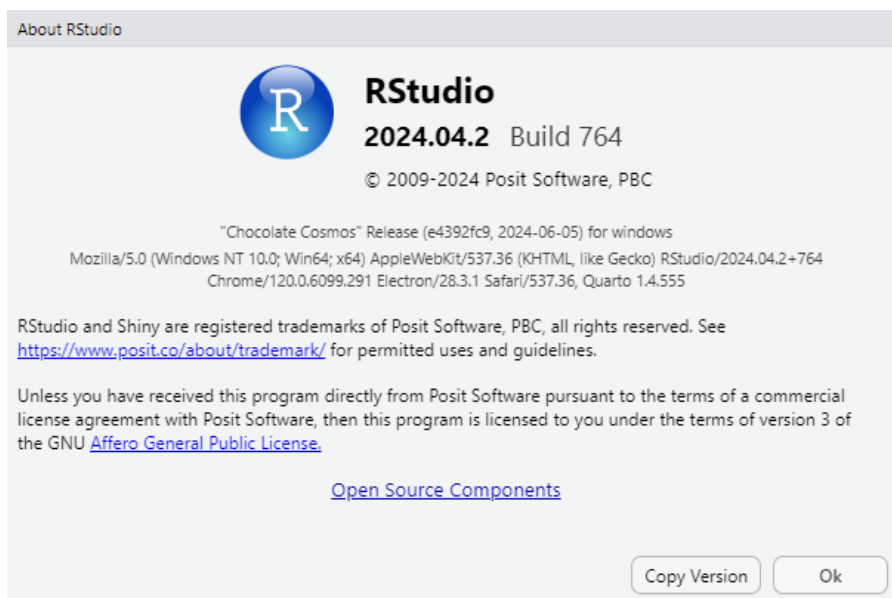
## Screenshot of R version

```
> R.version
```

```
platform      x86_64-w64-mingw32
arch           x86_64
os             mingw32
crt            ucrt
system        x86_64, mingw32
status
major          4
minor          3.3
year           2024
month          02
day            29
svn rev        86002
language       R
version.string  R version 4.3.3 (2024-02-29 ucrt)
nickname       Angel Food Cake
> |
```



## Screenshot of R Studio IDE version



# Analytics Programming Week-1 Solutions

2024-08-28

```
#Solution 1: creating numeric, character, logical and date data type
numeric_variable <- 11
char_variable <- 'a'
logical_variable <- TRUE
as.Date('2024-08-11')->date_variable

#validating the data type
print(paste("First data is", class(numeric_variable), "type."))

## [1] "First data is numeric type."

print(paste("Second data is", class(char_variable), "type."))

## [1] "Second data is character type."

print(paste("Third data is", class(logical_variable), "type."))

## [1] "Third data is logical type."

print(paste("Fourth data is", class(date_variable), "type."))

## [1] "Fourth data is Date type."

#Solution 2: vector of the names of the days of the week.
monthsOfYEAR <-
c("Sunday","Monday","Tuesday","Wednesday","Thursday","Friday","Saturday")

print(monthsOfYEAR)

## [1] "Sunday"      "Monday"      "Tuesday"     "Wednesday"   "Thursday"    "Friday"
## [7] "Saturday"
```

```
#Solution 3: vector of numbers that ranges from 200 to 220
dataVec1 <- 200:220
print(paste("Vector of number from 200 to 220 is: ",dataVec1))

## [1] "Vector of number from 200 to 220 is: 200"
## [2] "Vector of number from 200 to 220 is: 201"
## [3] "Vector of number from 200 to 220 is: 202"
## [4] "Vector of number from 200 to 220 is: 203"
## [5] "Vector of number from 200 to 220 is: 204"
## [6] "Vector of number from 200 to 220 is: 205"
## [7] "Vector of number from 200 to 220 is: 206"
## [8] "Vector of number from 200 to 220 is: 207"
## [9] "Vector of number from 200 to 220 is: 208"
## [10] "Vector of number from 200 to 220 is: 209"
## [11] "Vector of number from 200 to 220 is: 210"
## [12] "Vector of number from 200 to 220 is: 211"
```

```
## [13] "Vector of number from 200 to 220 is: 212"
## [14] "Vector of number from 200 to 220 is: 213"
## [15] "Vector of number from 200 to 220 is: 214"
## [16] "Vector of number from 200 to 220 is: 215"
## [17] "Vector of number from 200 to 220 is: 216"
## [18] "Vector of number from 200 to 220 is: 217"
## [19] "Vector of number from 200 to 220 is: 218"
## [20] "Vector of number from 200 to 220 is: 219"
## [21] "Vector of number from 200 to 220 is: 220"
```

*# Solution 4: Vecotr of numbers that ranges from 40 to 60*

```
dataVec2 <- seq(from = 40, to = 60, by = 1)
print(dataVec2)
```

```
## [1] 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
```

*# Solution 5: adding two vector*

```
dataCombVec1 <- c(dataVec1, dataVec2)
print(dataCombVec1)
```

```
## [1] 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216
217 218
## [20] 219 220 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54
55 56
## [39] 57 58 59 60
```

*# Solution 6: multiplying two vector*

```
datacombVec2 <- dataVec1 * dataVec2
print(datacombVec2)
```

```
## [1] 8000 8241 8484 8729 8976 9225 9476 9729 9984 10241 10500
10761
## [13] 11024 11289 11556 11825 12096 12369 12644 12921 13200
```

*#Solution 7*

*# NULL -> it represents the absence of value or missing value*

*#NA -> means undefined data within the data. You can remove NA using `NA.rm` to use sum command*

*# Solution 8: code that will install dplyr package in RStudio*

```
install.packages('dplyr')
```

```
library(dplyr)
```

```
packageVersion('dplyr')
```

```
## [1] '1.1.4'
```

```
summary(cars)
```

```
##      speed      dist
##  Min.   : 4.0   Min.   : 2.00
```

```
## 1st Qu.:12.0 1st Qu.: 26.00
## Median :15.0 Median : 36.00
## Mean :15.4 Mean : 42.98
## 3rd Qu.:19.0 3rd Qu.: 56.00
## Max. :25.0 Max. :120.00
```

## Including Plots

You can also embed plots, for example:

